Large language models predict human sensory judgments across six modalities

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Determining the extent to which the perceptual world can be recovered from language is a longstanding problem in philosophy and cognitive science. We show that state-of-the-art large language models can unlock new insights into this problem by providing a lower bound on the amount of perceptual information that can be extracted from language. Specifically, we elicit pairwise similarity judgments from GPT models across six psychophysical datasets. We show that the judgments are significantly correlated with human data across all domains, recovering well-known representations like the color wheel and pitch spiral. Surprisingly, we find that a model (GPT-4) co-trained on vision and language does not necessarily lead to improvements specific to the visual modality. To study the influence of specific languages on perception, we also apply the models to a multilingual color-naming task. We find that GPT-4 replicates cross-linguistic variation in English and Russian illuminating the interaction of language and perception.

perception | representation | NLP | AI | cognitive science

Imagine that you were chosen to be part of an expedition aimed at studying a newly discovered alien species on a distant planet. Your task is to understand the perceptual system of that species. You arrive at the planet and, to your dismay, discover that its inhabitants are long departed and the only thing they left behind is a huge archive of text. How much of the perceptual world of that species can you recover based on text alone? Versions of this question have occupied philosophers for centuries (1, 2), and decades of psychological research are beginning to provide glimpses into the rich perceptual content of language and its influence on perception (3–11).

But how can the amount of information that language provides about perception be quantified? Here we propose to do so by eliciting psychophysical judgments from large language models (LLMs) such as GPT-3 and its recent "ChatGPT" variants GPT-3.5 and GPT-4 (12, 13). These models are trained on massive text corpora reflecting a substantial chunk of human language and can be queried in a way that is analogous to humans.

Recent research shows that LLMs can be used to study various aspects of cognition (14), including, language processing in the brain (15–17), perception (18–20), cross-modal alignment (21), and morality (22, 23). Here, we use the fact that they are trained on large amounts of language to gain insight into the classic problem of the relationship between language and perception: these models provide a lower bound on the amount of information about perceptual experience that can be extracted from language alone.

We explored this empirically across six modalities, namely, pitch, loudness, colors, consonants, taste, and timbre. Given a stimulus space (e.g., colors) and its stimulus specification (e.g., wavelengths or their corresponding hex-codes) we elicit pairwise similarity judgments in a direct analogy to the widespread paradigm of similarity in cognitive science (24) using a carefully crafted prompt that is given to the model to complete (Figure 1A; see Methods). Importantly, whereas four of the modalities were based on classical results from the literature (colors (25), loudness (26), timbre (27) and taste (28)), two human datasets (pitch and vocal consonants) were novel and thus were not part of the training set of the models.

It is worthwhile to note that, unlike the other models, GPT-4 was trained in a multi-modal approach, enabling it to access both written text (similar to the other two variants) and images. This allowed us to examine if the additional sensory information resulted in enhanced performance in the color modality relative to the other domains. Moreover, to further interrogate whether the LLMs' sensory representation was language-dependent, we tested whether they would behave differently in the presence of the same sensory information (a color hex-code), but respond in different languages. To that end, we conducted a color-naming task using a paradigm similar to that of (29) and the World Color Survey (30, 31), and constructed human and GPT-3, GPT-3.5, and GPT-4 color-naming maps in both English and Russian.

Results

Similarity study. For each dataset, we designed a tailored prompt template that could be filled in with in-context examples and the pair of target stimuli for which we would want the LLM to produce a similarity rating (see Methods and SI for full specification of the prompts and datasets). Across all domains, we elicited 10 ratings per pair of stimuli from each of the GPT models and then constructed aggregate similarity ratings by averaging. We then evaluated the resulting scores by correlating them with human data. The Pearson correlation coefficients between human data and model predictions are shown in Figure 1B (See SI for details regarding computing the correlations and CIs). We see that across all domains, the correlations were significant, and were particularly high for pitch ($r = .92$, 95% CI $[.91, .92]$ for GPT-4), loudness ($r = .89$, 95% CI $[.87, .91]$ for GPT-3.5), and colors ($r = .89$, 95% CI $[.87, .91]$ for GPT-4) (and $> .6$ for all models), followed by moderate but highly significant correlations for consonants ($r = .57$, 95% CI $[.56, .59]$ for GPT-4), taste ($r = .54$, 95% CI $[.48, .61]$ for GPT-3.5) and timbre ($r = .42$, 95% CI $[.40, .44]$

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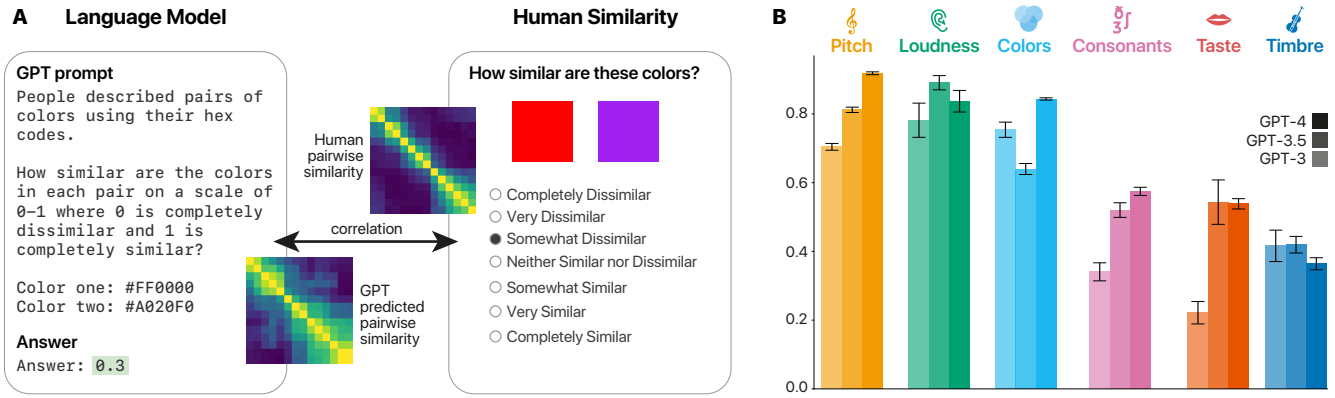

Fig. 1. A. Schematic of the LLM-based and human similarity judgment elicitation paradigms. **B.** Correlations between models and human data across six perceptual modalities, namely, pitch, loudness, colors, consonants, taste, and timbre (Pearson r; 95% CIs).

for GPT-3.5). For the two modalities for which we collected data, we could compare model performance to the inter-rater split-half reliability (IRR). The IRRs for pitch and consonants were $r = .90$ (95% CI $[.87, .92]$) and $r = .46$ (95% CI $[.36, .56]$), respectively, suggesting that the performance of GPT-4 is on par with human performance.

We also note that in five out of the six domains, GPT-4 was among the top two models. Interestingly, the improvement relative to the average performance of the other models happened across all modalities with the exception of timbre and loudness, and was not restricted or particularly large for the domain of colors (compare e.g. $\Delta r = .15$ 95% CI $[.13, .16]$ for colors vs. $\Delta r = .16$ 95% CI $[.15, .17]$ for pitch, $\Delta r = .16$ for taste 95% CI $[.12, .19]$, and $\Delta r = .14$ 95% CI $[.12, .16]$ for consonants) suggesting that this improvement is driven by richer textual training in GPT-4 rather than the possibility of its inclusion of images in its training set as is currently being debated (18).

Next, to get a finer picture of the LLM-based judgments and see to what extent they reflect human representations, we performed the following analyses. Starting from the domain of pitch, we wanted to see to what extent the LLM data captures a well-known psychological phenomenon that Western listeners tend to associate particular musical intervals or ratios of frequencies (such as the octave or 2:1 frequency ratio) with enhanced similarity (32). To test this we computed the average similarity score over groups of pitch pairs that are separated by the same fixed interval (i.e., the same frequency ratios). Figure 2A shows the resulting average similarity per interval for the models and humans along with an example corresponding smoothed similarity matrix for GPT-3 (smoothing was done by averaging the raw similarity matrix over its sub-diagonals). We can see that apart from the decay as a function of separation (i.e., tones that are far apart in log frequency are perceived as increasingly dissimilar) there is a clear spike precisely at 12 semitones (octave), consistent with the aforementioned phenomenon of "octave equivalence" (33). Moreover, applying multi-dimensional scaling (MDS) (24) to the smoothed similarity matrix whereby the different stimuli are mapped into points in a Euclidean space (also known as "psychological space") such that similar stimuli are mapped to nearby points reveals a clear helical structure with twists that correspond to precisely 12 semitone separations (i.e., octaves) recovering the pitch spiral representation (Figure 2A). Likewise, applying MDS to the domains of consonants and colors (Figure 2B) reveals highly interpretable representations, namely, the familiar color wheel and a production-based representation for consonants.* As an additional test, we asked GPT-4 to provide explanations for the judgments it made (Figure 2C-D), and remarkably, the model resorted to explanations involving the octave, ratios, and harmonic relations for pitch, places of articulation in the vocal tract for consonants, and hue, brightness and color spectra for colors, consistent with the MDS solutions.

Color naming study. The results so far suggest that LLMs can use textual information to form perceptual representations. If this is indeed the case, we hypothesized that representations in LLMs using different languages could be different even in the presence of identical input. This would be consistent with cross-cultural differences in language and perception that were observed in humans (30). To test this, we propose for the first time to test LLMs on an explicit naming task proposed by the seminal work of (29) and further explored across cultures around the globe (30, 31).

We thus tested whether LLMs would yield different naming patterns of color hex codes depending on the language of the prompt used to elicit those names (see Methods; Figure 3). Specifically, we presented both humans and LLMs with different colors and asked them to perform a forced-choice naming task by selecting from a pre-specified list of 15 color names (see Methods). We specifically focused on English and Russian as test cases, since Russian speakers are documented to use richer vocabulary to describe what English speakers would otherwise describe as blue and purple (29, 34, 35). We collected data from 103 native English speakers and 51 native Russian speakers and compared them against LLMs performing the same task, and to the in-lab data of (31) as an additional baseline.†

The results are shown in Figure 3. Our first finding was that GPT-4 in both English and Russian were more human-like than the other variants when compared with an adjusted Rand index (see Figure 3A; English: GPT-4 0.59 95% CI $[0.56, 0.63]$, GPT-3.5, 0.50 95% CI $[0.46, 0.52]$, GPT-3 0.39 95% CI $[0.37, 0.42]$;

* An interactive visualization of all human similarity matrices and their corresponding LLM counterparts, as well as their two-dimensional MDS solutions is available at: https://computational-audition.github.io/LLM-psychophysics/all-modalities.html

† The human color naming data that we collected can be explored via: https://computational-audition.github.io/LLM-psychophysics/color.html

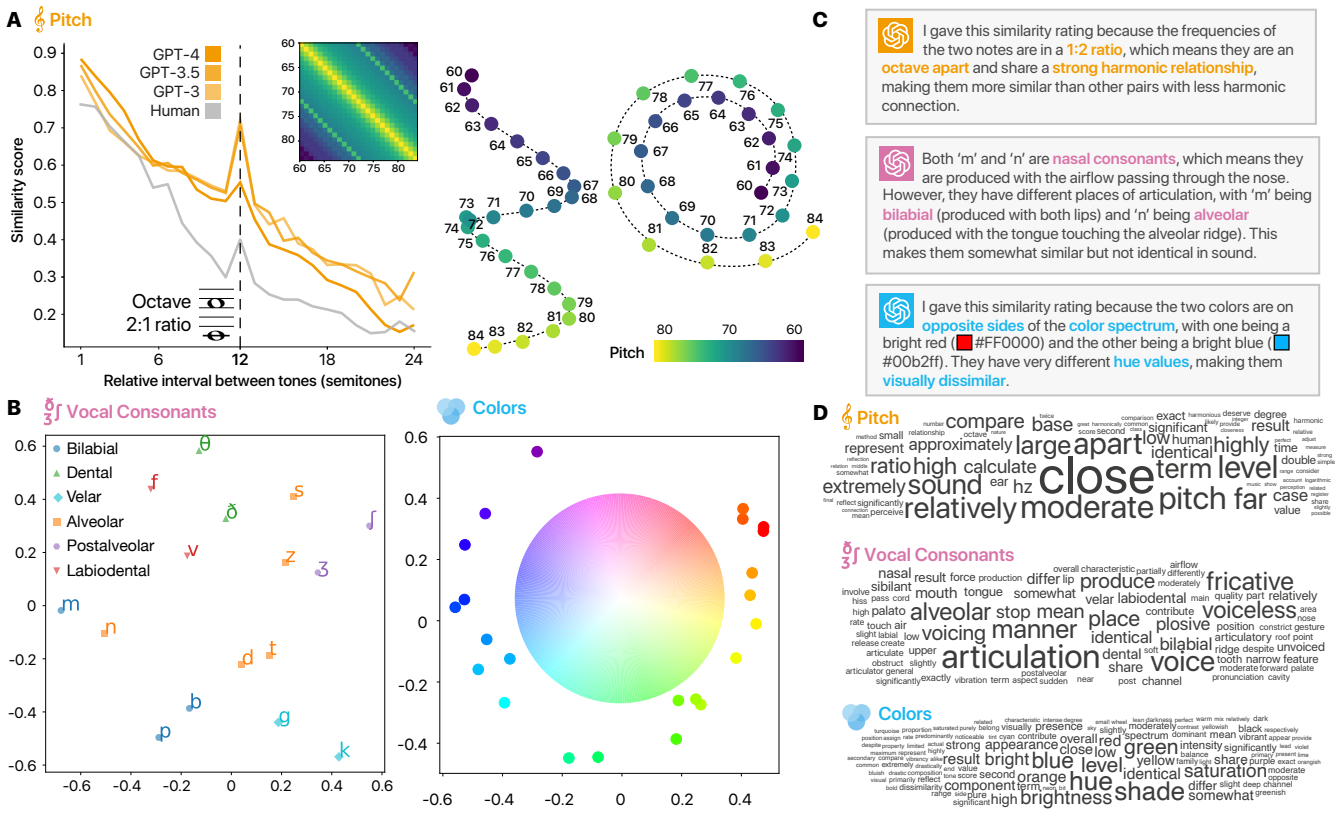

Fig. 2. A. Human and LLM similarity marginals and an example GPT-3 corresponding similarity matrix and its three-dimensional MDS solution for pitch. **B.** MDS solutions for vocal consonants and colors for GPT-4 similarity matrices. To illustrate the structure of the results, we highlighted consonants with the same place of articulation in the vocal tract with the same shape and color, and added a rotated HSV color wheel for the color MDS. **C.** Example GPT-4 explanations for similarity judgment scores. **D.** Word clouds for GPT-4 explanations in the domain of pitch, vocal consonants, and colors.

Russian: GPT-4 0.54 95% CI [0.46, 0.54], GPT-3.5 0.50 95% CI [0.45, 0.52], GPT-3 0.35 95% CI [0.29, 0.35]; see Methods). It is evident from our data, however, that LLMs are still not perfect in predicting human color naming as compared to a separate lab-based experiment conducted by Lindsey and Brown (31) (dashed line in Figure 3A top, constrained naming task 0.73 95% CI [0.65,0.73], free naming task 0.75 95% CI [0.66,0.75]). Moreover, the naming of GPT-4 colors differs from human data in some important cases, including the color turquoise, which was selected as the dominant color for 46 Munsell colors in GPT4 versus only 15 in human data. Note, however, that our human English results conducted online and with relative less control over color presentation were highly consistent with (31) that were conducted in the lab and under controlled environment, even though (31) used slightly different paradigms: free naming (consistency to our human data: 0.75 95% CI [0.66, 0.75]) and forced-choice list with a different set of items (0.73 95% CI [0.65, 0.74]).

Importantly, however, GPT-4 appears to replicate cross-lingual differences (Figure 3B), for example separating Russian blue and purple into distinct categories for lighter and darker areas (35). Indeed, the color sínij / Синий (Blue) was the dominant category for 18 and 29 Munsell colors for GPT-4 and humans, respectively, and the color golubój / Голубой (Light-blue) was accordingly the dominant category for 33 and 26 colors. Similarly, the color fiolétovyj / Фиолетовый (Violet) was the dominant category for 27 and 32 Munsell colors for GPT-4 and humans, respectively, and the color lilóvyj / Лиловый (Lilac) was accordingly the dominant category for 20 and 18 Munsell colors.

Discussion

In this work, we showed how recent advances in large language models and, in particular, their flexible prompt engineering capabilities provide an elegant way for extracting clear quantitative psychophysical signals from text corpora. Our results contribute to a variety of thought-provoking issues in perception and language research. In particular, our findings further support recent research suggesting that people that lack direct sensory experiences (e.g., congenitally blind individuals) could still possess a rich understanding of perceptual concepts through language (e.g., colors (10)). Likewise, the language-dependence of the color representation of GPT-4 in the naming task suggests that the physical stimulus alone (in our case, the color hex code) is insufficient for explaining the behavioral patterns of the model since the same hex codes triggered different color categorizations in Russian and English. This supports the idea that language is not only shaped by perception but can also reshape it (35). Finally, the fact that GPT-4 achieved IRR-level performance without any fine-tuning in the newly collected datasets of pitch and vocal consonants, contributes to our understanding of LLMs' ability to mimic human behavior (13).

We end by discussing some limitations which point towards